Circumference, arcs and perimeter consolidation

Day 17 • 2026-11-27 • 10:30 AM–12:00 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

Which formula finds circumference when diameter d is known?

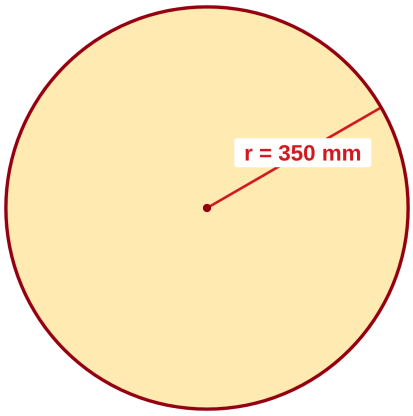
A. C = pi × d²
B. C = d ÷ pi
C. C = 2 × d²
D. C = pi × d

Choice (A–D) _____ **Working / reason** _____

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02

The Shape Canvas circle has radius 350 mm. Find its diameter.



Shape Canvas

Answer (mm) _____ **Working / reason** _____

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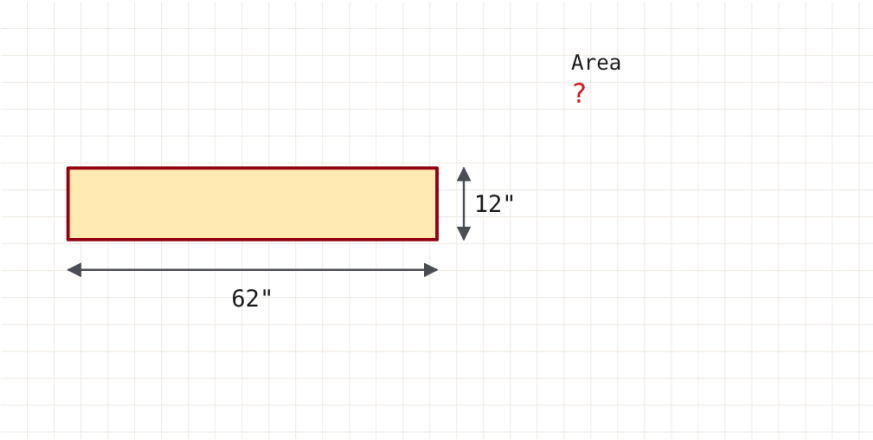
03 Consider a semicircular panel, separate from the Numeracy Hub rectangle shown. Which pieces make the panel’s full perimeter?

A. The full circumference plus radius

B. Half the circumference plus the diameter

C. Half the circumference only

D. The diameter only



Numeracy Hub

Choice (A–D)

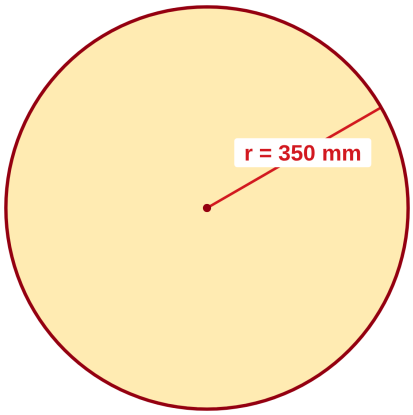
Working / reason

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04 Find the circumference of the Shape Canvas circle with radius 350 mm. Use pi = 3.14.



Shape Canvas

Answer (mm)

Working / reason

Textbook sections 9.2–9.6 pp.222–229

05 What fraction of a circle’s circumference is a 90° arc?

A. One half

B. Three quarters

C. A full circle

D. One quarter

Choice (A–D)

Working / reason

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06 CALCULATE

A circle has diameter 0.8 m. Find its circumference using pi = 3.14.

Answer (m) | **Working / reason**

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07 A circular opening has radius 125 mm. Find its circumference using pi = 3.14.

Answer (mm) | **Working / reason**

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08 Find the length of a 90° arc with radius 2 m. Use pi = 3.14.

Answer (m) | **Working / reason**

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09 Find the full perimeter of a semicircular panel with radius 1.5 m. Use pi = 3.14.

Answer (m) | **Working / reason**

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10 A 120° arc has radius 450 mm. Find its length using pi = 3.14.

Answer (mm)

Working / reason

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11 APPLY & CHECK

A student uses C = pi × r when r is the radius. What is missing?

A. A square on r

B. Division by 360

C. The radius must be halved

D. A factor of 2

Choice (A–D)

Working / reason

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12 A curved boundary is a semicircle of radius 2 m joined to two straight sides of 3 m each. Find the total of those three boundary pieces. Use pi = 3.14.

Answer (m)

Working / reason

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13 A calculation gives circumference 50 mm for a circle 200 mm in diameter. Which estimate reveals the problem?

A. Circumference should be 40000 mm²

B. Circumference should be a little over 600 mm

C. Circumference should be less than the diameter

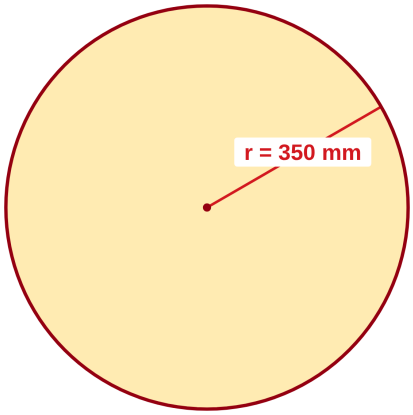
D. Circumference should equal 100 mm

Choice (A–D)

Working / reason

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14 A semicircular trim has radius 350 mm as in Shape Canvas. Find only its curved length using pi = 3.14.



Shape Canvas

Answer (mm)

Working / reason

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15 When calculating an arc length, why is angle/360 included?

A. It changes millimetres to metres

B. It finds the radius

C. It squares the diameter

D. It gives the fraction of the full circle

Choice (A–D)

Working / reason

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